

Challenges (2): data storage and access

Creating a global viral kinetics library

A library of standardized viral kinetics data with basic model fits/parameter estimates

Complementary to efforts like:

- WHO Global Repository of Epidemic Parameters
- Global Vaccine Library
- NextStrain

Benefits:

- Facilitates comparisons between study designs, pathogens, time frames, demographic groups...
- Enables quick dissemination of data
- Enables closer collaboration across the data collection/modeling/policy design spectrum

Challenges:

- Balancing data comparability with completeness
- Ensuring data quality
- Integrating robust + flexible modeling tools
- Ensuring appropriate credit (see next slide!)

The screenshot shows the 'Filters' section of the Viral Kinetics Explorer. It includes an 'Age Range' slider set from 0 to 100, a 'Study ID' field with 'ke2022' and 'kissler2023' selected, and a 'Sample Type' field with 'nasal', 'saliva', and 'nasal_orophary...' selected.

Viral Kinetics Explorer

Upload your viral kinetics CSV (or use default below)

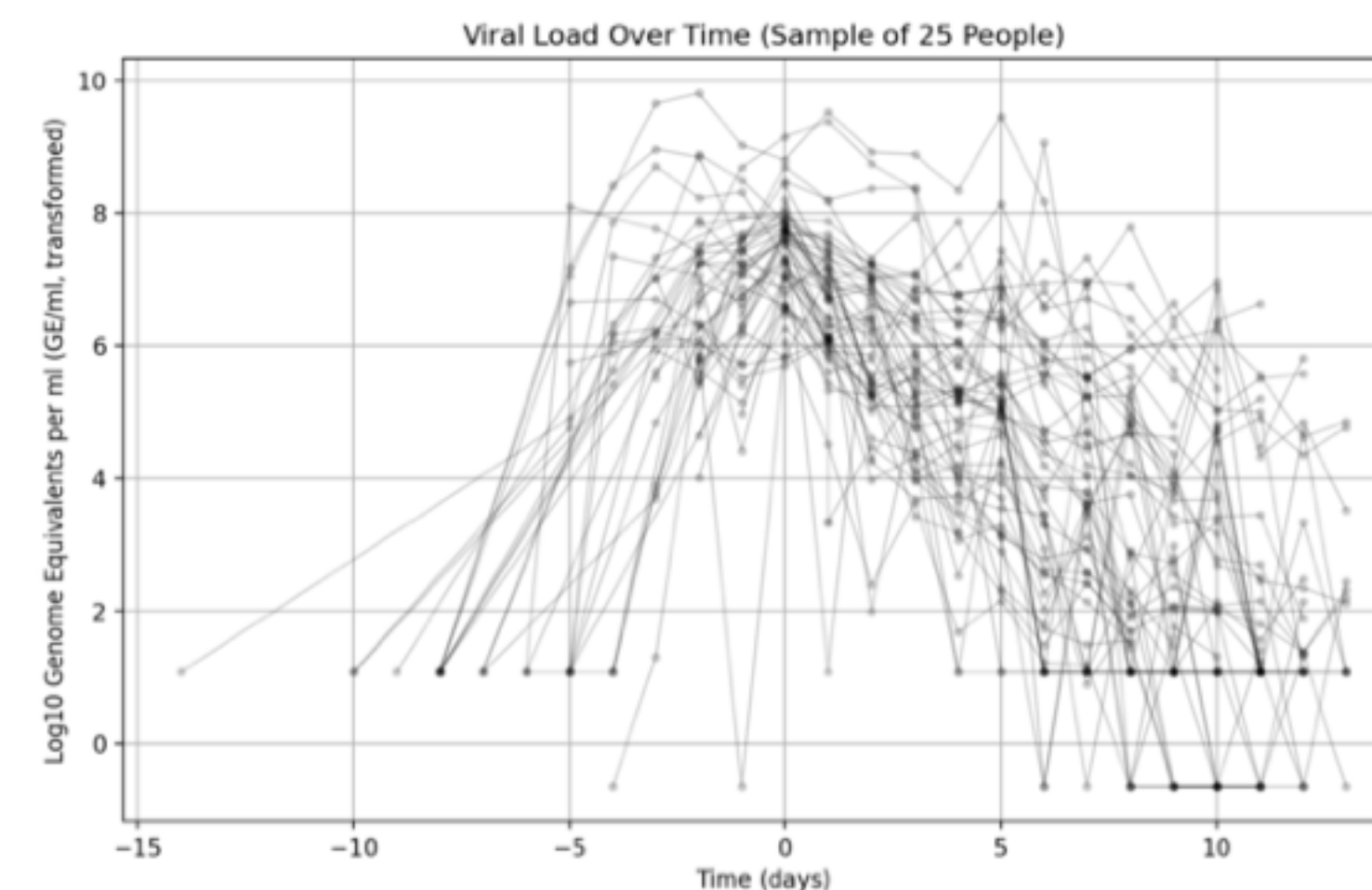
Drag and drop file here
Limit 200MB per file • CSV

Browse files

Using default viral kinetics dataset.

Filtered Data Preview

	StudyID	PersonID	InfectionID	SampleID	TimeDays	Symptoms1	Symptoms2	Symptoms3	Symptoms4	Comorbidity1	Comorbidity2	Comorbidity3	Comorbidity4	T
2	ke2022	443108	None	None	-2	None	None	None	None	None	None	None	None	
3	ke2022	443108	None	None	-1	None	None	None	None	None	None	None	None	
4	ke2022	443108	None	None	0	None	None	None	None	None	None	None	None	
5	ke2022	443108	None	None	1	None	None	None	None	None	None	None	None	
6	ke2022	443108	None	None	2	None	None	None	None	None	None	None	None	
7	ke2022	443108	None	None	3	None	None	None	None	None	None	None	None	
8	ke2022	443108	None	None	4	None	None	None	None	None	None	None	None	
9	ke2022	443108	None	None	5	None	None	None	None	None	None	None	None	
10	ke2022	443108	None	None	6	None	None	None	None	None	None	None	None	
11	ke2022	443108	None	None	7	None	None	None	None	None	None	None	None	



Challenges (3): data reporting and sharing standards

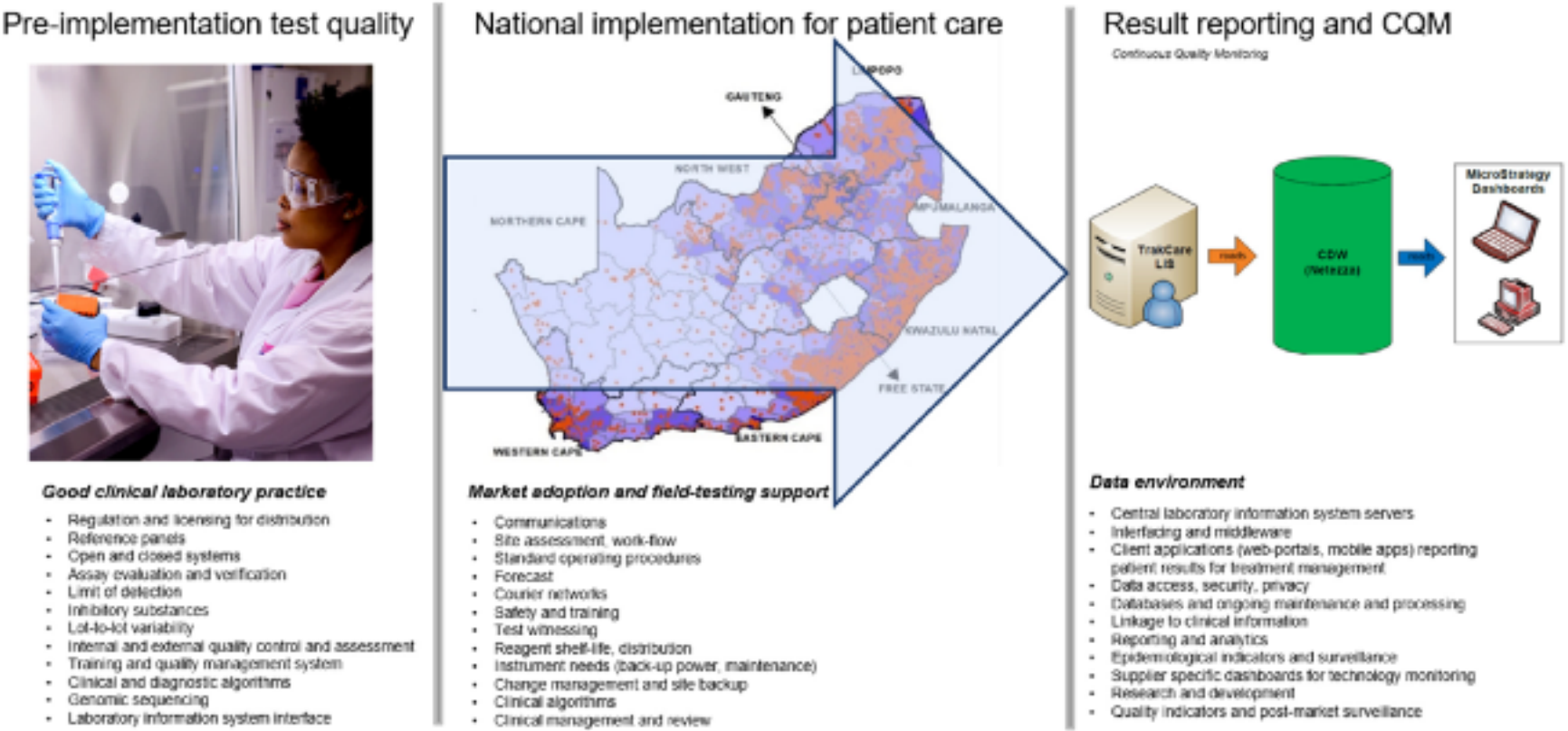
Establishing standards for robust and ethical data reporting/sharing

Nationwide Harmonization Effort for Semi-Quantitative Reporting of SARS-CoV-2 PCR Test Results in Belgium

by Lize Cuypers ^{1,*}, Jannes Bode ¹, Kurt Beuselinck ¹, Lies Laenen ¹, Klaas Dewaele ¹, Reile Janssen ¹, Arnaud Capron ², Yves Lafort ², Henry Paridaens ³, Bertrand Bearzatto ⁴, Mathieu Cauchie ⁵, Aline Huwart ⁶, Jonathan Degosserie ⁷, Olivier Fagnart ⁸, Yarah Overmeire ⁹, Arlette Roufflange ¹⁰, Ilse Vandecandelaere ¹¹, Marine Deffontaine ¹², Thomas Pilate ¹³, Nicolas Yin ¹⁴, Isabel Micalessi ¹⁵, Sandrine Roisin ¹⁶, Veronique Moons ¹⁷, Marijke Reynders ¹⁸, Sophia Steyaert ¹⁹, Coralie Henin ²⁰, Elena Lazarova ²¹, Dagmar Obbels ²², François E. Dufrasne ²³, Hendri Pirenne ²⁴, Raf Schepers ²⁵, Anaëlle Collin ²⁶, Bruno Verhasselt ²⁷, Laurent Gillet ²⁸, Stijn Jonckheere ²⁹, Philippe Van Lint ³⁰, Bea Van den Poel ³¹, Yollen Van der Beken ³², Violeta Stojkovic ³³, Maria-Grazia Garrino ³⁴, Hannah Segers ³⁵, Kevin Vos ³⁶, Maaike Godefroid ³⁷, Valerie Pede ³⁸, Friedel Nollet ³⁹, Vincent Claes ⁴⁰, Inge Verschraegen ⁴¹, Pierre Bogaerts ⁴², Marjan Van Gysel ⁴³, Judith Leurs ⁴⁴, Veroniek Saegeman ⁴⁵, Oriane Soetens ⁴⁶, Merijn Vanhee ⁴⁷, Gilberte Schlettekatte ⁴⁸, Evelyne Huyghe ⁴⁹, Steven Martens ⁵⁰, Ann Lemmens ⁵¹, Heleen Nailis ⁵², Kim Laffineur ⁵³, Deborah Steensels ⁵⁴, Elke Vanlaere ⁵⁵, Jérémie Gras ⁵⁶, Gatién Roussel ⁵⁷, Koenraad Gijbels ⁵⁸, Michael Boudewijns ⁵⁹, Catherine Sion ⁶⁰, Wim Achtergaël ⁶¹, Wim Maurissen ⁶², Luc Iliano ⁶³, Marianne Chantrenne ⁶⁴, Geert Vanheule ⁶⁵, Reinoud Flies ⁶⁶, Nicolas Hougardy ⁶⁷, Mario Berth ⁶⁸, Vanessa Verbeke ⁶⁹, Robin Morent ⁷⁰, Anne Vankeerberghen ⁷¹, Sébastien Bontems ⁷², Kaat Kehoe ⁷³, Anneleen Schallier ⁷⁴, Giang Ho ⁷⁵, Kristof Bafort ⁷⁶, Marijke Raymaekers ⁷⁷, Yolande Pypen ⁷⁸, Amelie Heinrichs ⁷⁹, Wim Schuermans ⁸⁰, Dominique Cuigniez ⁸¹, Salah Eddine Lali ⁸², Stefanie Drieghe ⁸³, Dieter Ory ⁸⁴, Marie Le Mercier ⁸⁵, Kristel Van Laethem ^{86,87}, Inge Thoelen ⁸⁸, Sarah Vandamme ⁸⁹, Iqbal Mansoor ⁹⁰, Carl Vael ⁹¹, Maxime De Sloovere ⁹², Katrien Declerck ⁹³, Elisabeth Dequeker ¹, Stefanie Desmet ^{1,94}, Piet Maes ⁸⁷, Katrien Lagrou ^{1,94} and Emmanuel André ^{1,86,94} — Hide full author list

How South Africa Used National Cycle Threshold (Ct) Values to Continuously Monitor SARS-CoV-2 Laboratory Test Quality

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FAIR Principles

GO FAIR is committed to making data and services **findable, accessible, interoperable and reusable (FAIR)**.



Findable: Metadata and data should be easy to find for both humans and computers.



Accessible: The exact conditions under which the data is accessible should be provided in such a way that humans and machines can understand them.



Interoperable: The (meta)data should be based on standardized vocabularies, ontologies, thesauri etc. so that it integrates with existing applications or workflows.



Reusable: Metadata and data should be well-described so that they can be replicated and/or combined in different research settings.

Table 1. Proposed interpretation and reporting scheme for SARS-CoV-2 PCR results [24]: a semi-quantitative reporting approach of PCR test results into four categories of SARS-CoV-2 positivity.

Category of SARS-CoV-2 Positivity	SARS-CoV-2 Viral Load (RNA Copies/mL)	Interpretation with Respect to Infectivity
Very strongly positive	≥10 ⁷	patient is contagious
Strongly positive	≥10 ⁵ –<10 ⁷	patient is probably contagious
Moderate positive	≥10 ³ –<10 ⁵	patient is potentially contagious, unless there is clinical and/or serological evidence of an old, cleared infection
Weakly positive	<10 ³	patient is probably not or no longer contagious if there is also clinical and/or serological evidence of an old, cleared infection